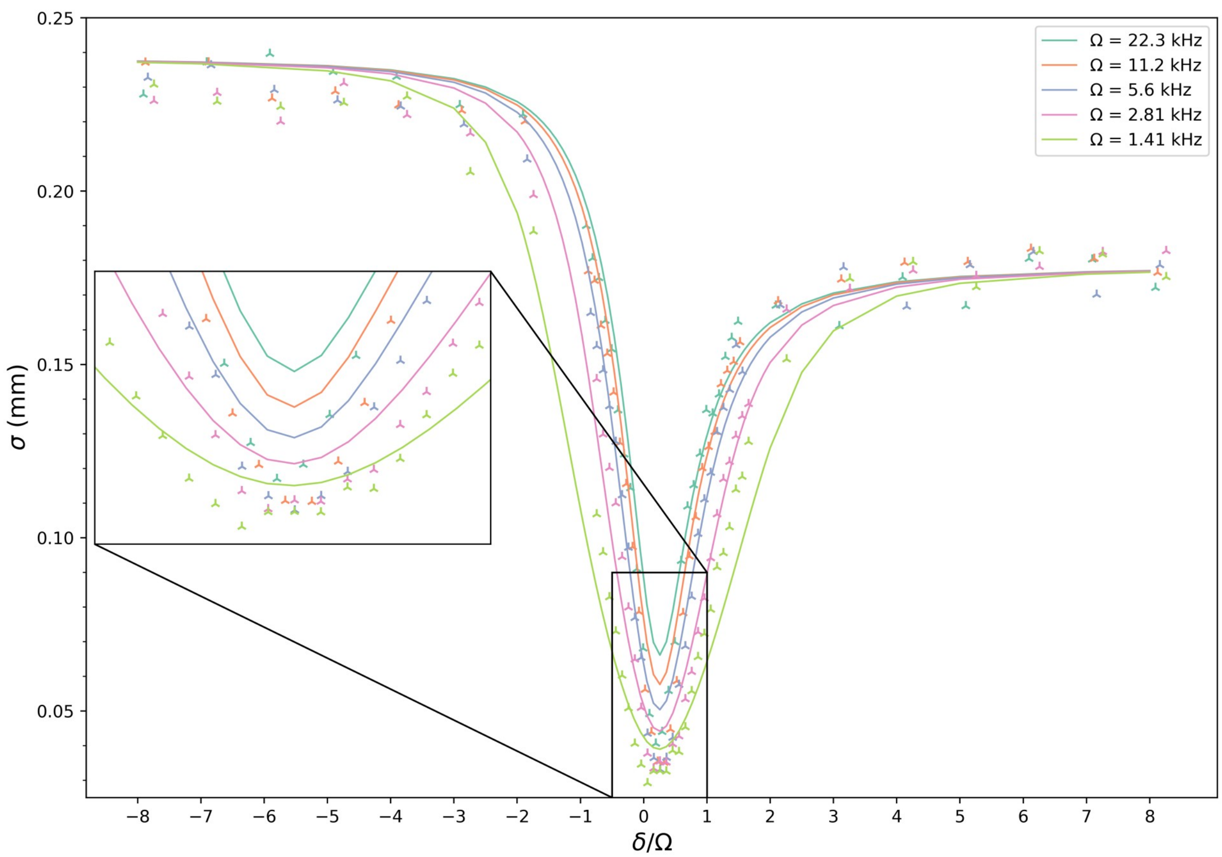
Report GPELab simulations

Simulations with BMF effects

Simulations with BMF effects added in modifying a_{12} accordingly in GPE simulation setting parameters

In [10]:
```
from IPython.display import display, Image

fileName = r"C:\Users\sarat\OneDrive\Documenti\GitHub\Numerical-simulations\GPELab\
display(Image(filename=fileName))
```

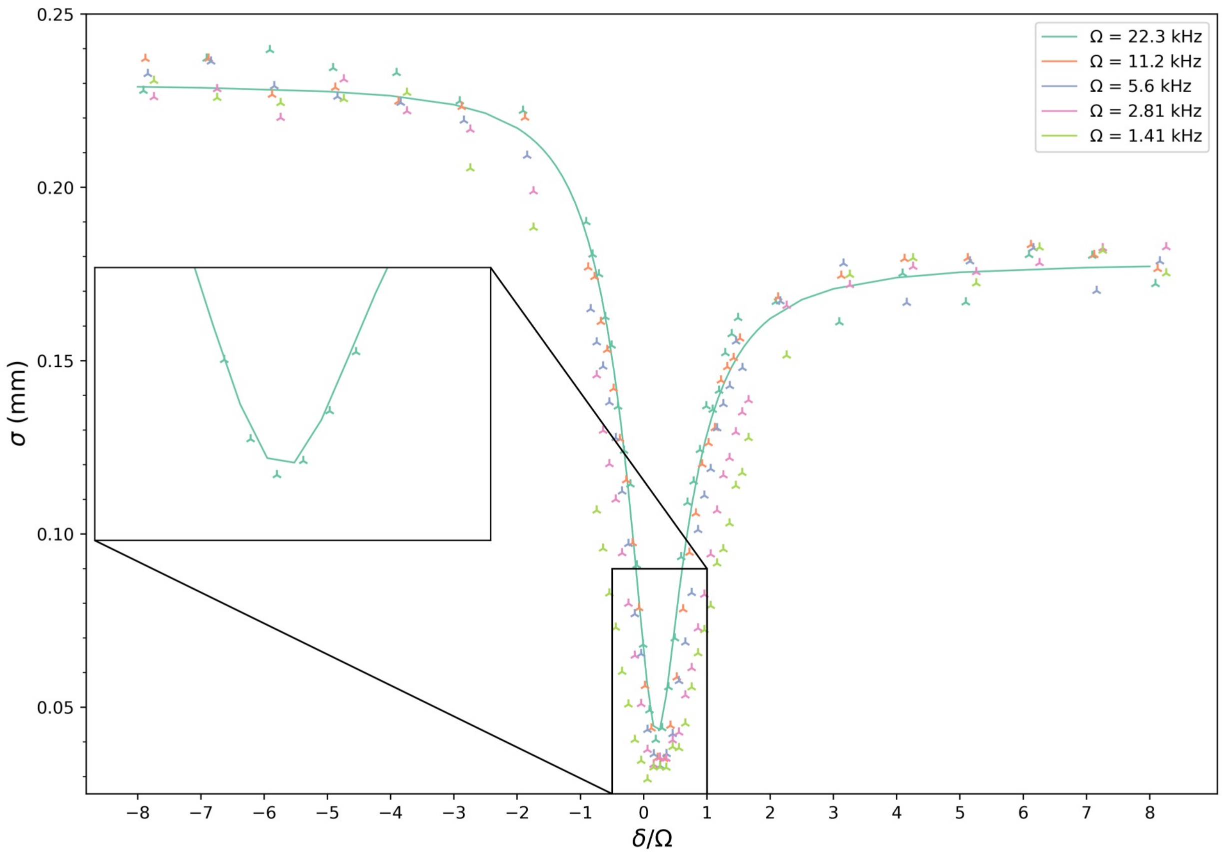


COMMENTS ON "simulations with BMF"

BMF effects added in the simulation modifying a12
We can appreciate how probably the experimental data do not capture as much BMF effect as we thought. At least we clearly see that in the simulations the BMF shift af delta=0 is larger then the shift in the data. In fact we can calculate min point for all the curves and then evaluate the difference between the minima for the 22kHz curve and for the 1.4kHz curve.
The result is:
SIMULATION $\Delta = 0.027$ mm
EXP DATA $\Delta = 0.0114$ mm

I could try to adjust the scattering length a11 and the density so to fit well the 22kHz curve. And then evaluate all the other curves for those parameters.
In ordet to fit the 22kHz cirve I needed to lower $a_{11} \to a_{11} - 10a_0$ and to higher the peak density $n_0 = 5.3e9$. The curve is so fitted well (see plot below). The problem is that for those parameters the lower Rabi curves can not be simulated because we so in the attractive region. This is the proof that the data do not have as much BMF effects. In addiction I know that for negative detuning the numper of atoms is lower than at the beginning of the sweep, so I expect the simulaiton to be abowe the data in this region.

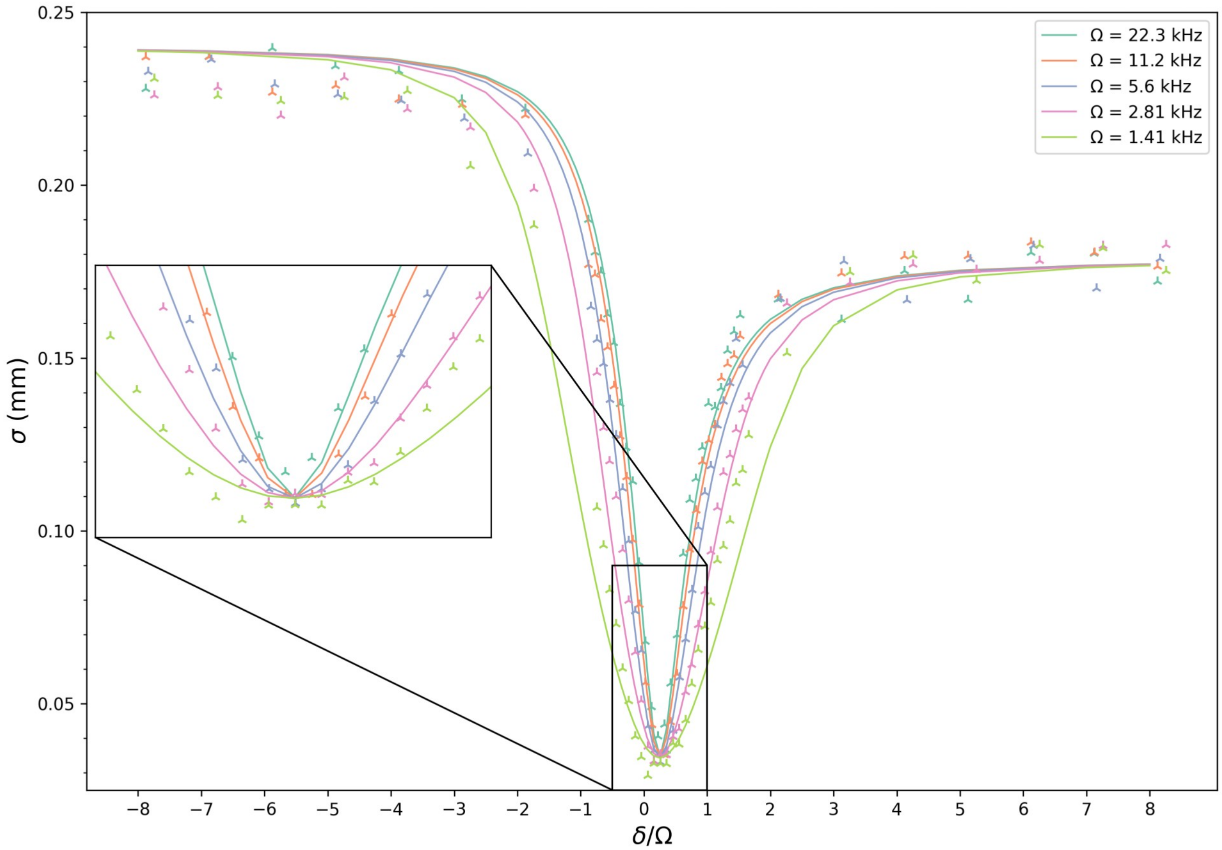
In [13]:
```
fileName = r"C:\Users\sarat\OneDrive\Documenti\GitHub\Numerical-simulations\GPELab\
display(Image(filename=fileName))
```



MF simulations: adjustment of Tiemann scattering lengths

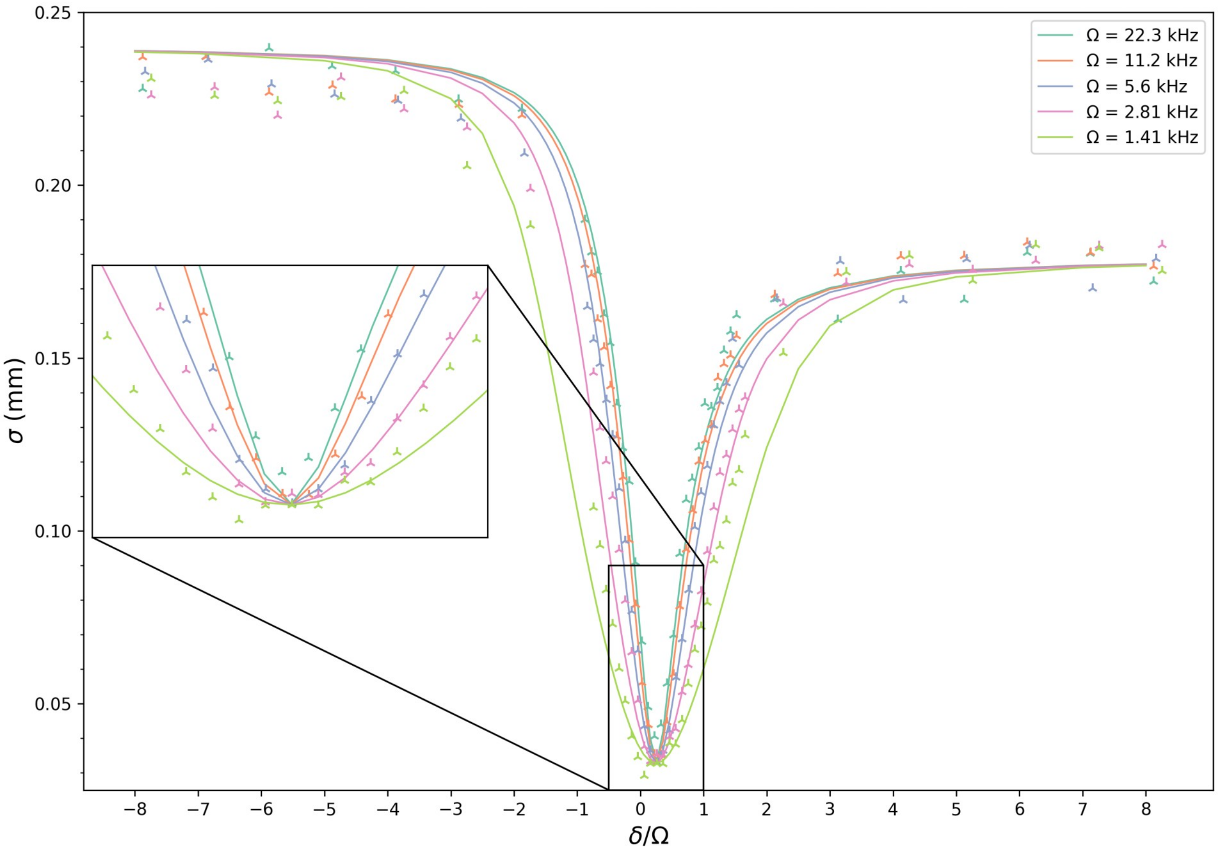
If we run the simulation setting the Tiemann scattering lengths the min size of the curves is always (a lot) below the data. This we have now seen is not (mostly) because of BMF effects. We have so probably to adjust the a11 scattering length to higher the min points of the simulated sizes.
I put a11 + 2

In [16]:
```
fileName = r"C:\Users\sarat\OneDrive\Documenti\GitHub\Numerical-simulations\GPELab\
display(Image(filename=fileName))
```



I repeat the simulation but that tomi I set $a_{11} \to a_{11} + 1.65$. I am trying to fit better the min for the 1.4 kHz curve (that one with less MF effects). Then I simulate the rest of the surves with the same paramenters. I note that the 1.4kHz curve in not fitting well the 3 body...

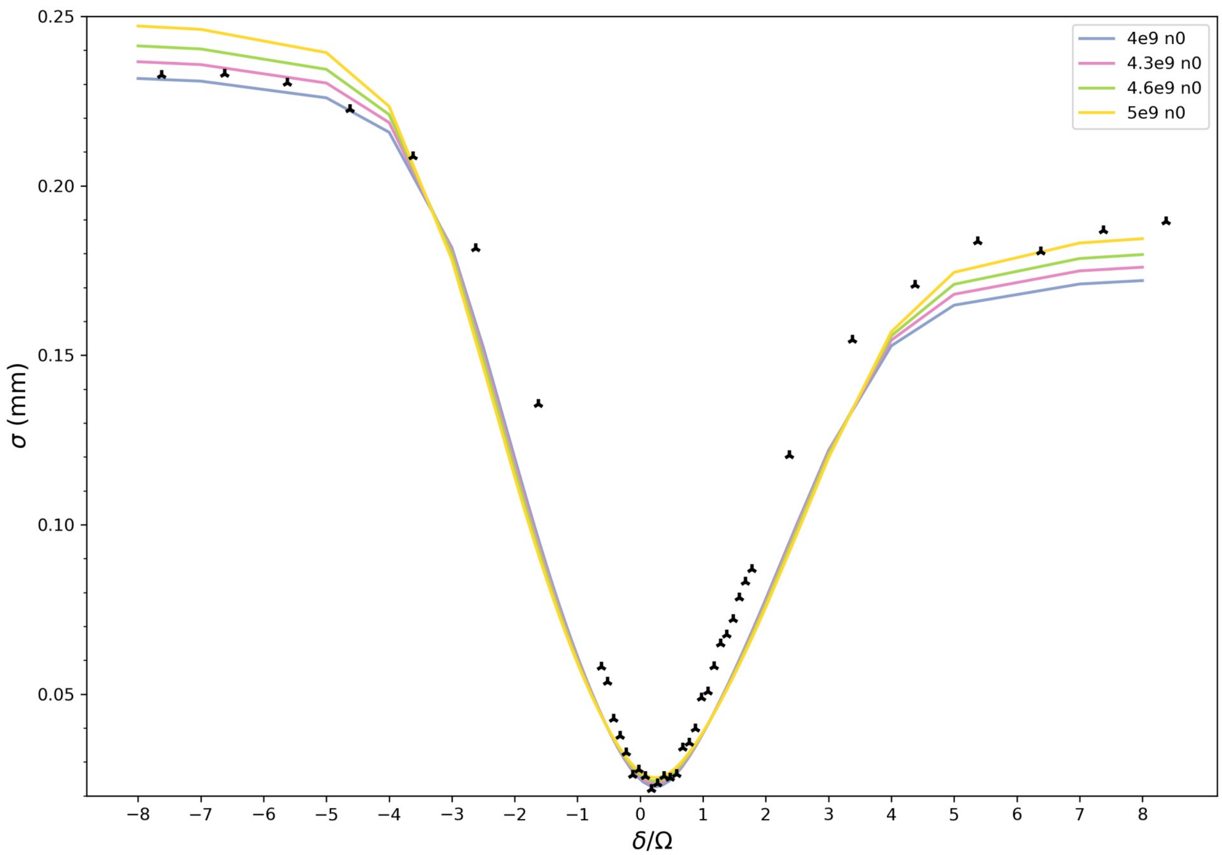
In [19]:
```
fileName = r"C:\Users\sarat\OneDrive\Documenti\GitHub\Numerical-simulations\GPELab\
display(Image(filename=fileName))
```



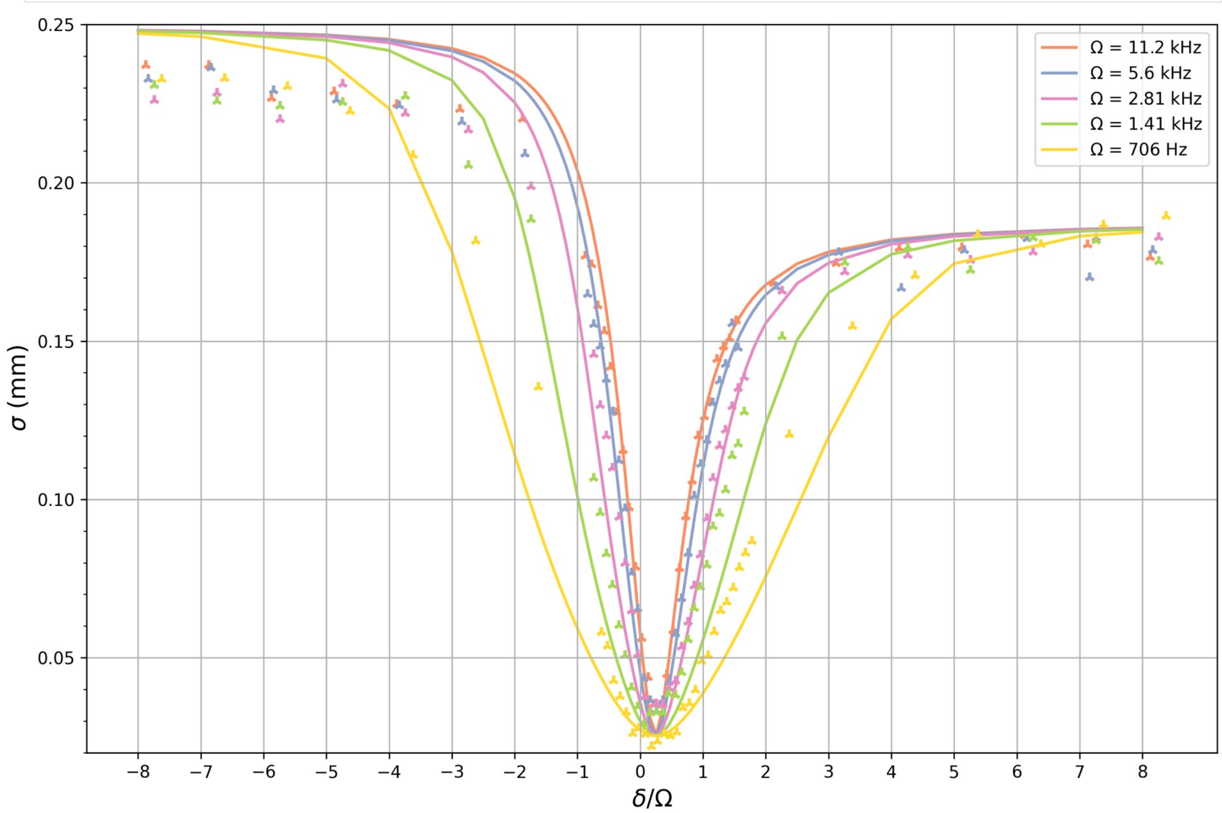
MF simulation: Tiemman scatteing lengths

Actually is one takes into account also the lowest curve ($\Omega = 700$ Hz) the MF simulation really fits the minimum of this curve... I now rum divverent simulations tuning the peak density to fit well the data at big positive detuning (at really low detuning the simulation will be probably higher than the data since we have some losses of atoms in the adiabatic sweep.

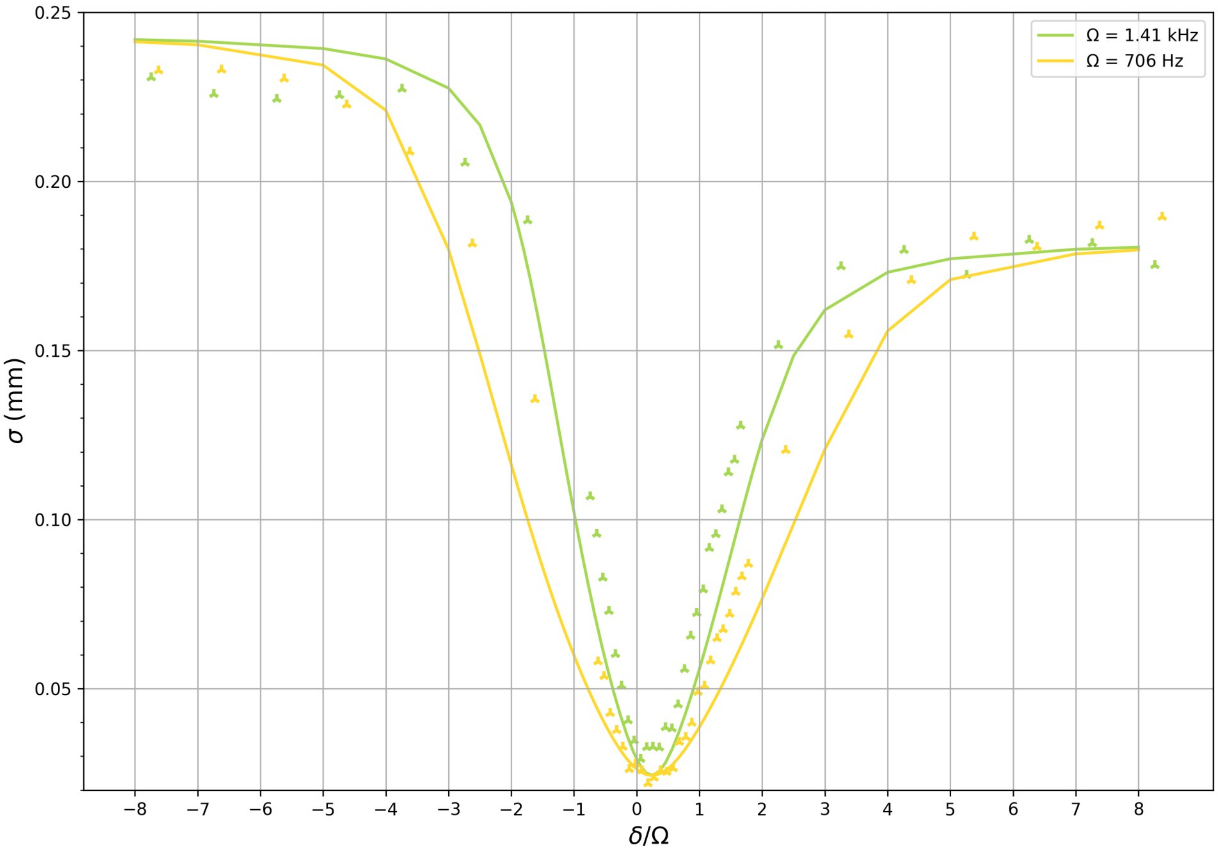
In [7]:
```
fileName = r"C:\Users\Sarah\Documents\GitHub\Numerical-simulations\GPELab\plot_code
display(Image(filename=fileName))
```



In [6]:
```
fileName = r"C:\Users\sarat\OneDrive\Documenti\GitHub\Numerical-simulations\GPELab\
display(Image(filename=fileName))
```



In [5]:
```
fileName = r"C:\Users\sarat\OneDrive\Documenti\GitHub\Numerical-simulations\GPELab\
display(Image(filename=fileName))
```


Ω = 1.41 kHz
Ω = 706 Hz
σ (mm)
δ/Ω
0.25
0.20
0.15
0.10
0.05
−8
−7
−6
−5
−4
−3
−2
−1
0
1
2
3
4
5
6
7
8

In [ ]: